

ECON 2: Principles of Macroeconomics

Lecture 11: Expenditure Multiplier and Fiscal Policy



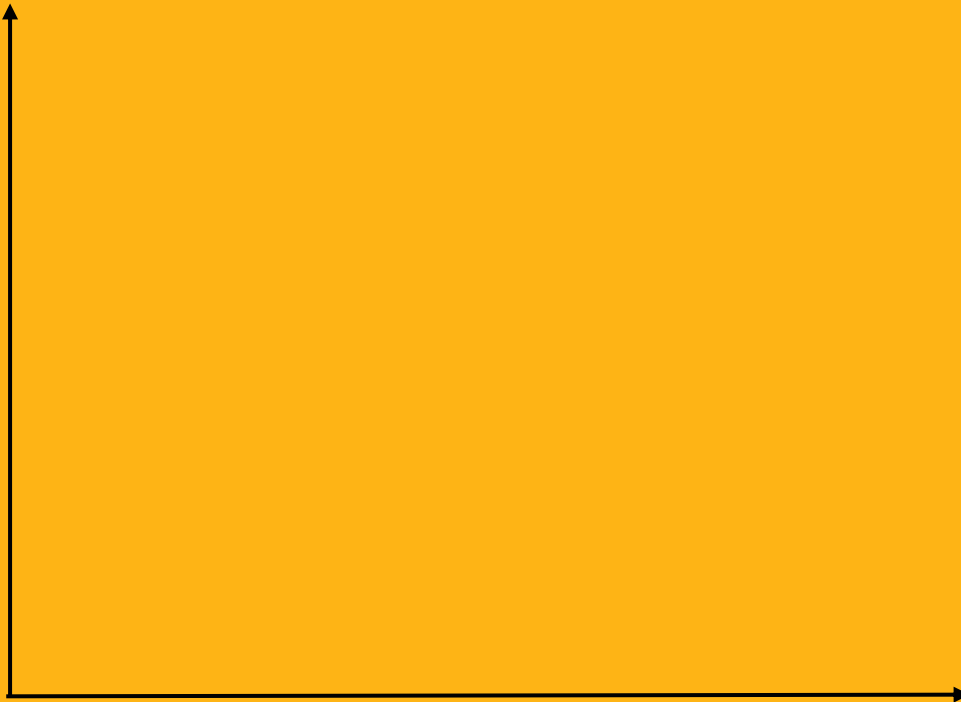
Today: Chapters 4.3 and 5.1

- Expenditure Multiplier
- Role of the Federal Government
- Expenditure vs.
Tax Multiplier



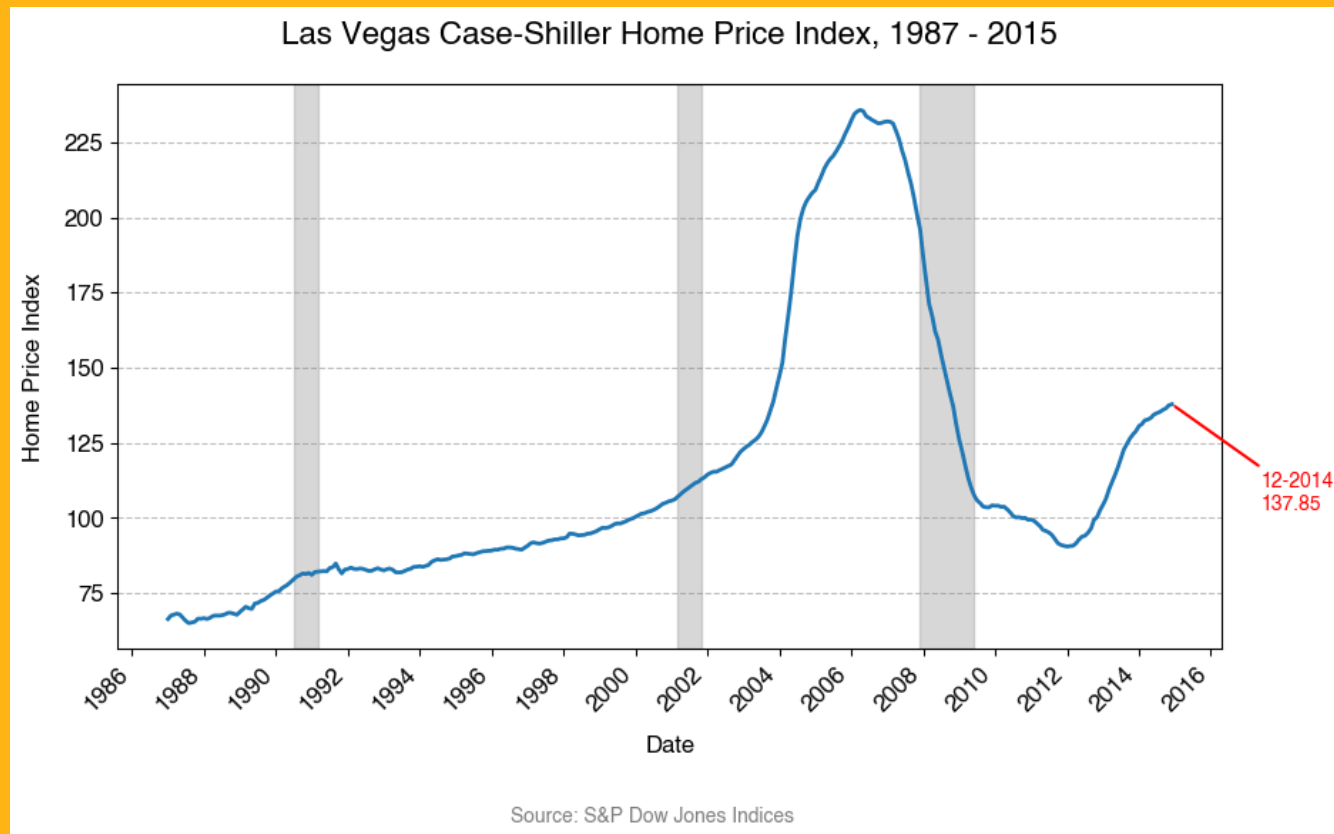
Changes in Equilibrium

- Start at $Y^* = 10,000$ (full-employment output)



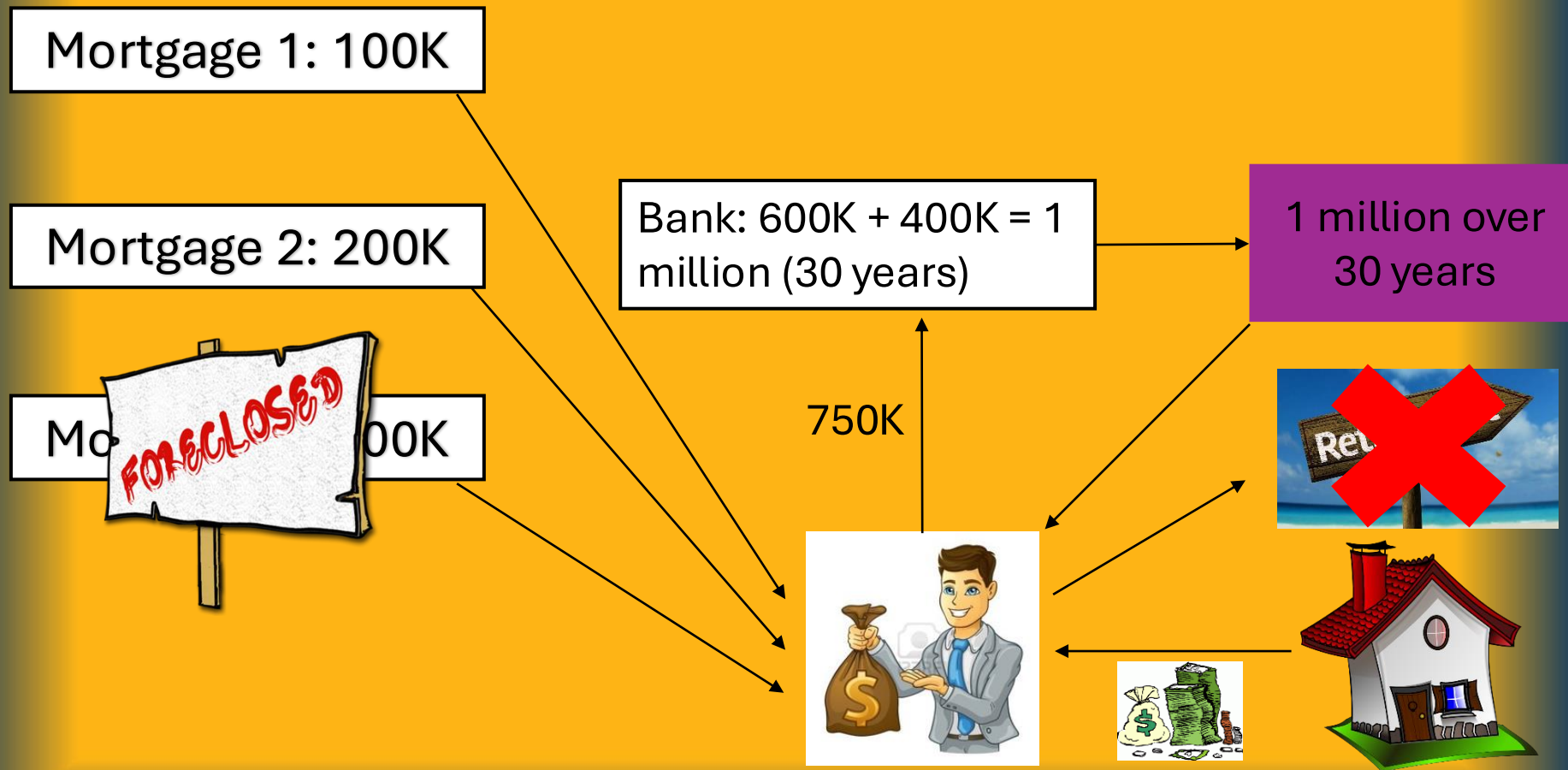
Housing Market Equilibrium

- Las Vegas Housing Prices Over Time

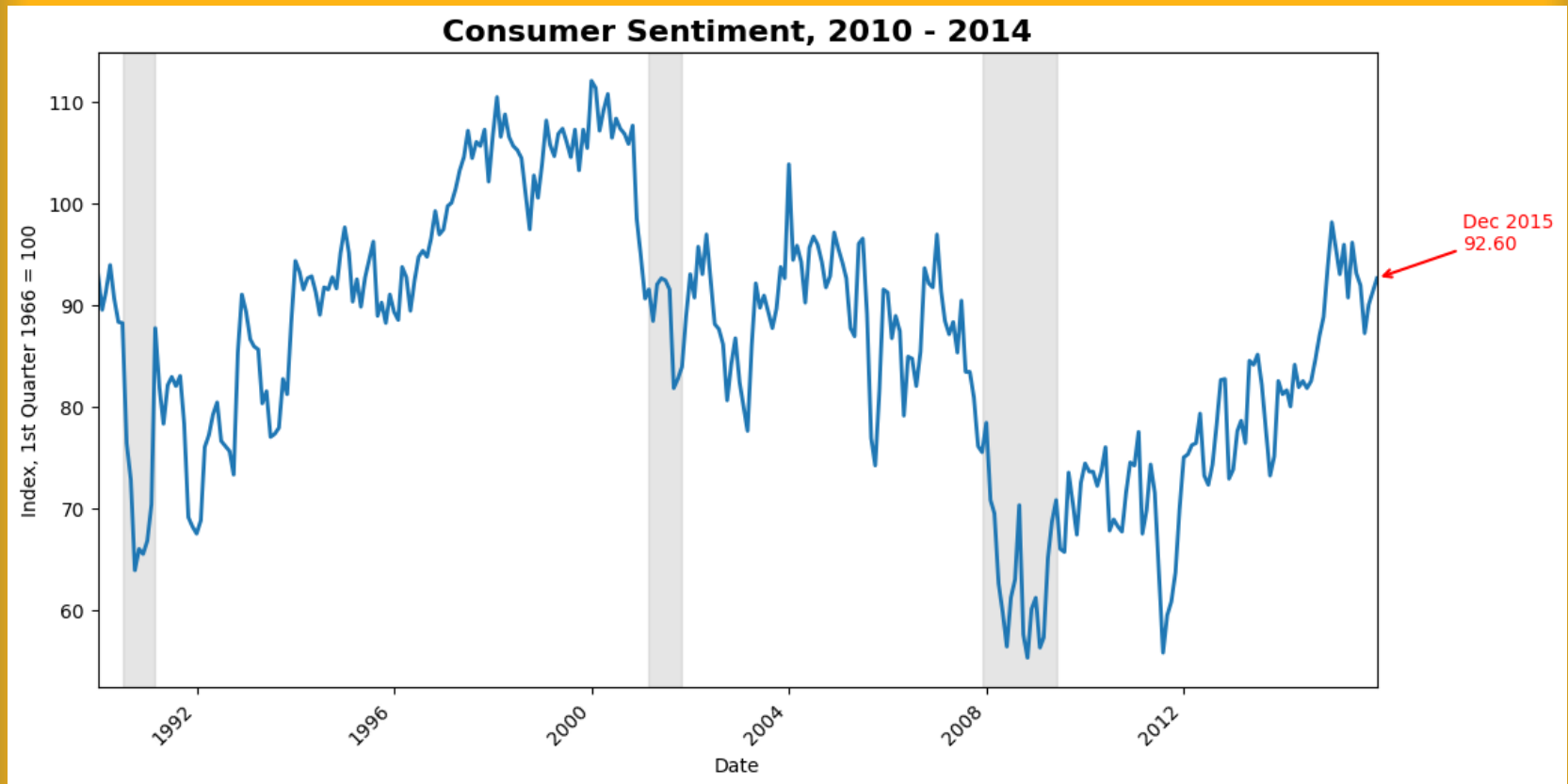


Consequences of the Housing Crash

- Mortgage Backed Securities

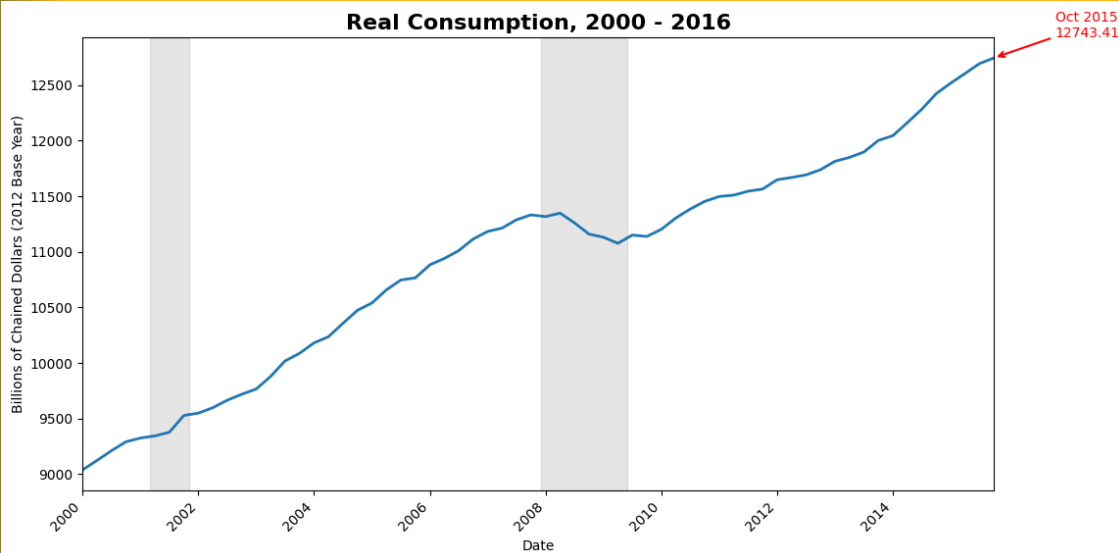


Consumer Confidence Decreased

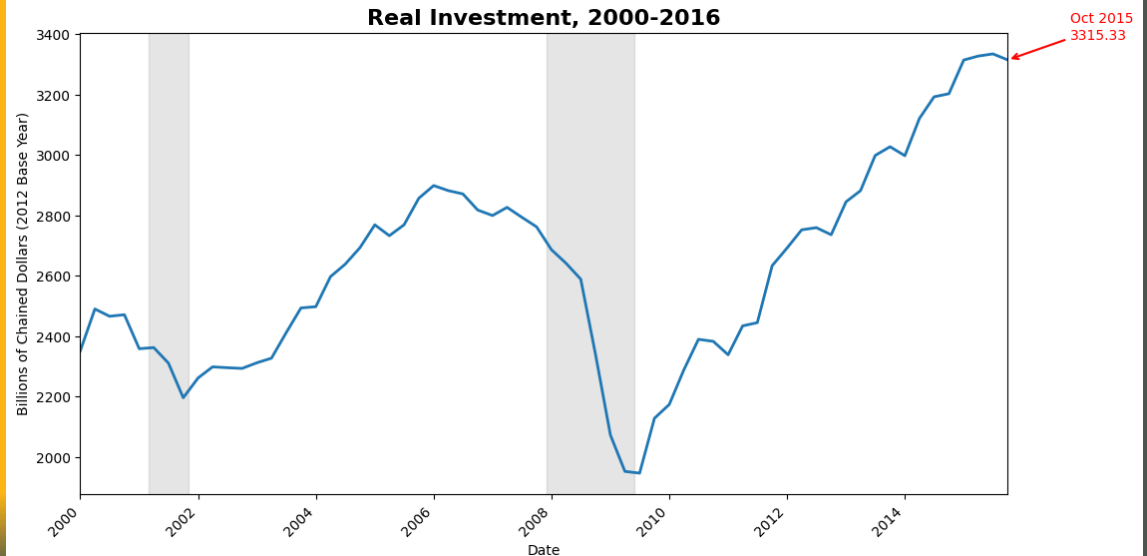


Housing: Decrease in AC and I

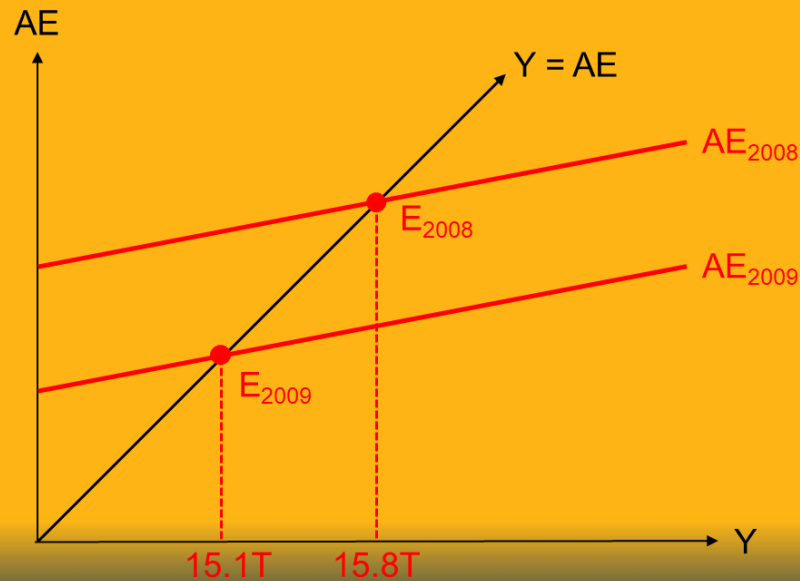
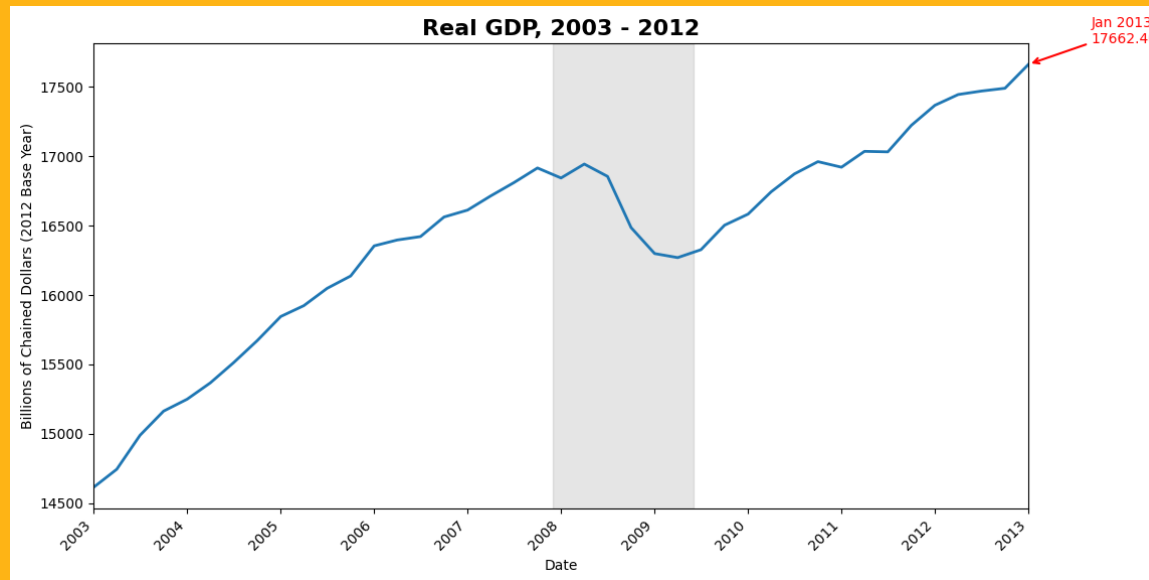
Real Consumption, 2000 - 2016



Real Investment, 2000-2016

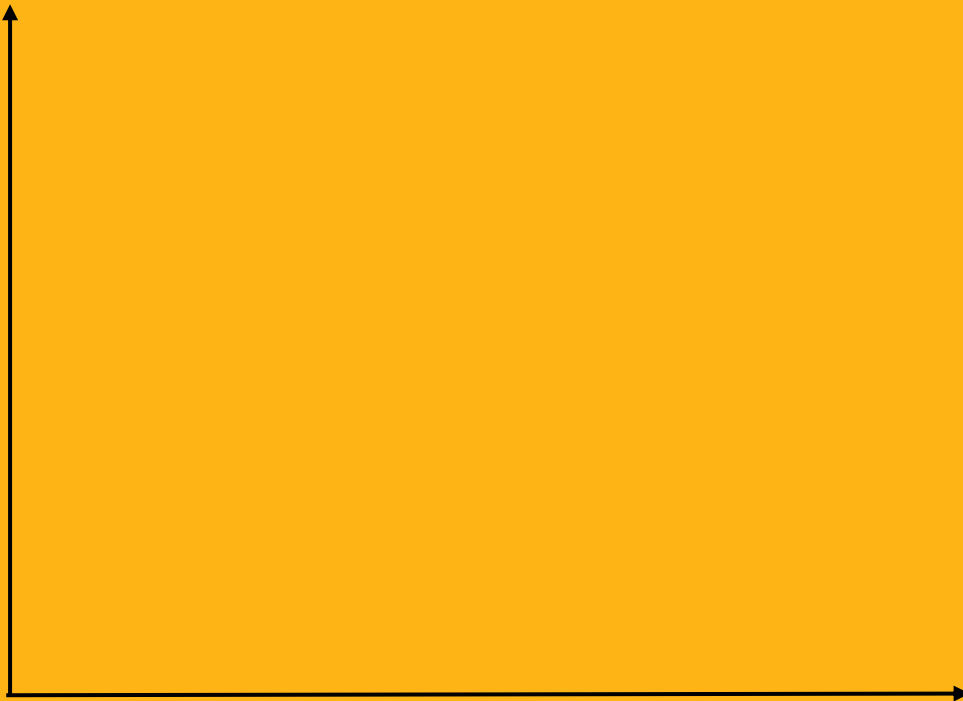


Great Recession



Changes in Equilibrium

- Start at $Y^* = 10,000$ (full-employment output)
Housing Crash! (assume $mpc = 0.8$)



Recovering from a Crash

- Option 1: Rich person buys \$500 worth of machines (I^P)

\$500 in machines = increase I^P by \$500
Incomes increase by \$500

Recovering from a Crash

- Option 1: Rich person buys \$500 worth of machines (I^P)

Round 1:
Y up \$500

\$500 in machines = increase I^P by \$500
Incomes increase by \$500

Round 2:
Y up \$400 more,
\$900 total

Spend $mpc \times 500 =$
 $0.80 \times 500 = \$400$
Consumption increase

Save $mps \times 500 =$
 $0.20 \times 500 = \$100$

Recovering from a Crash

- Option 1: Rich person buys \$500 worth of machines (I^P)

Round 1:
Y up \$500

\$500 in machines = increase I^P by \$500
Incomes increase by \$500

Round 2:
Y up \$400 more,
\$900 total

Spend $mpc \times 500 =$
 $0.80 \times 500 = \$400$
Consumption increase

Save $mps \times 500 =$
 $0.20 \times 500 = \$100$

Incomes rise by \$400

Recovering from a Crash

- Option 1: Rich person buys \$500 worth of machines (I^P)

Round 1:
Y up \$500

\$500 in machines = increase I^P by \$500
Incomes increase by \$500

Round 2:
Y up \$400 more,
\$900 total

Spend $mpc \times 500 =$
 $0.80 \times 500 = \$400$
Consumption increase

Save $mps \times 500 =$
 $0.20 \times 500 = \$100$

Incomes rise by \$400

Round 3:
Y up \$320,
\$1,220 total

Spend $mpc \times 400 =$
 $0.80 \times 400 = \$320$
Consumption increase

Save $mps \times 400 =$
 $0.20 \times 400 = \$80$

Recovering from a Crash

Round	Additional I ^P	Additional Consumption	Total Additional Spending = Additional GDP
1	500	0	500
2	0	$mpc \times 500 = 400$	$500 + 400 = 900$
3	0	$mpc \times 400 = 320$	$500 + 400 + 320 = 1,220$
4	0	$mpc \times 320 = 256$	$500 + 400 + 320 + 256 = 1,476$
5	0	$mpc \times 256 = 204.8$	$500 + 400 + 320 + 256 + 204.8 = 1680.8$
...			
10	0	$mpc \times 83.89 = 67.11$	2231.56
...			
25	0	$mpc \times 2.95 = 2.36$	2490.56
...			
50	0	$mpc \times 0.011 = 0.009$	2500

Recovering from a Crash

- Original Increase in I^P = \$500
- Total Increase in GDP = ?
- Expenditure Multiplier =

Calculating the Expenditure Multiplier

- How to figure out the total change in GDP when I^P increases by \$500?
- Total Additional GDP = $\$500 + mpc \times \$500 + mpc \times (mpc \times \$500) + \dots$
Round 1 + Round 2 + Round 3
- Re-write as: $\$500 \times (1 + mpc + mpc^2 + mpc^3 + \dots) = \$500 \times \frac{1}{1-mpc} = \$500 \times \frac{1}{mps}$
- Expenditure multiplier = $\frac{1}{1-mpc}$
- How to use the expenditure multiplier: Increase in I^P is \$500, $mpc = 0.8$
- Total Change in GDP = $\$500 \times \frac{1}{1-0.8} = \$500 \times 5 = \$2,500$
- GDP started at \$7,500, ended at \$10,000
- Same result if we change G, NX, AC!!!

Finding Equilibrium

- Economy:
- $I^P = 500$; $G = 200$; $NX = 40$

Real GDP (Y)	Consumption (C)	Planned Investment (I^P)	Government Expenditures (G)	Net Exports (NX)	Aggregate Expenditures (AE)
5000	4560	500	200	40	
6000	5460	500	200	40	
7000	6360	500	200	40	
8000		500	200	40	
9000		500	200	40	
10000		500	200	40	

- MPC?
- Where is $\bar{Y}$? How can we get there?

Role of the Federal Government

Executive (President) and Legislative
(Congress/Parliament) Branches of Government:

Move Y to full-employment Y!

“Federal Government” = “Government” =
“President/Congress”

What can the Government change that will
influence aggregate expenditures/equilibrium Y?

1.

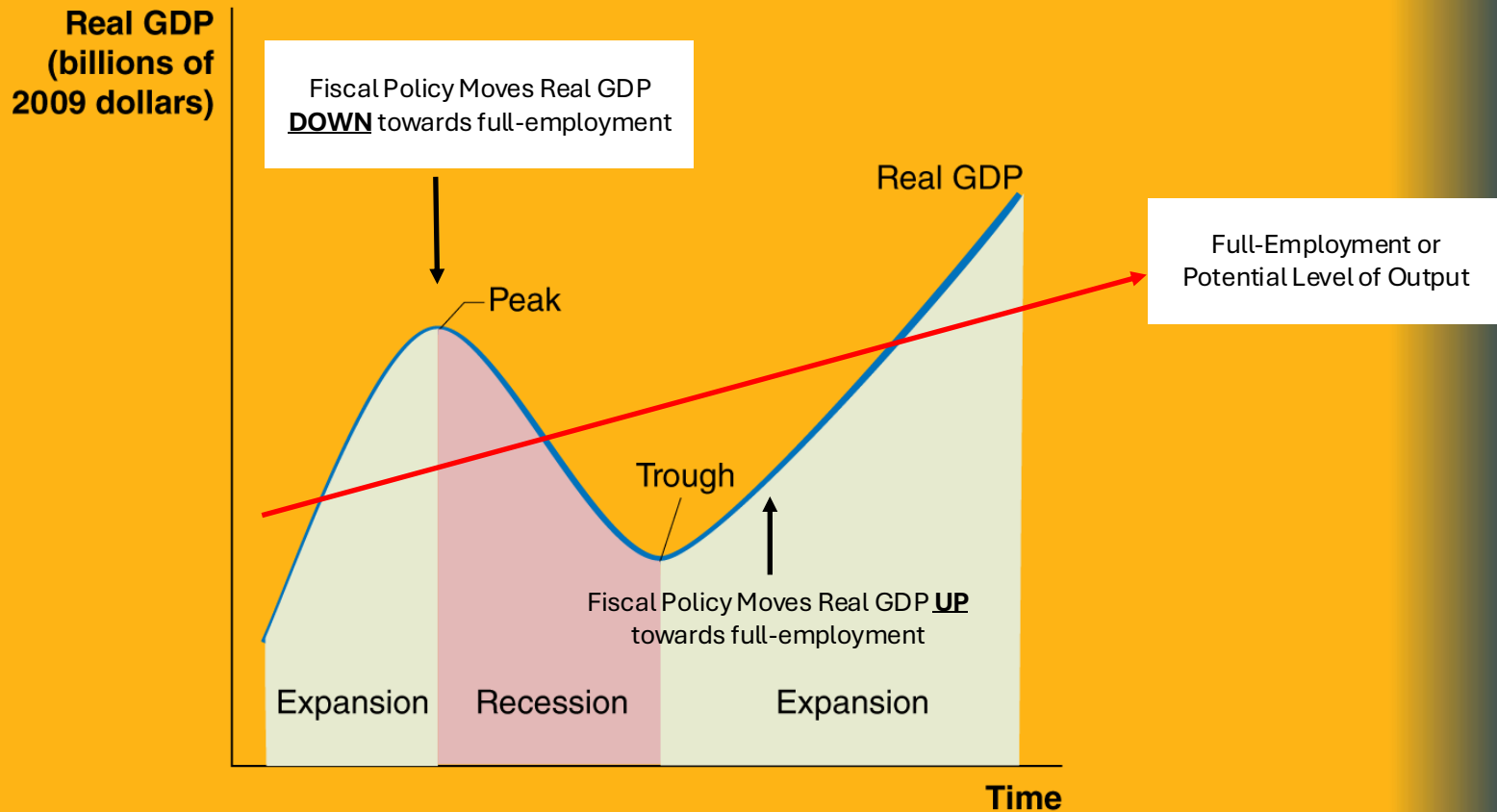
2.

Fiscal Policy:

Counter-Cyclical Fiscal Policy:



Counter-Cyclical Fiscal Policy



Fiscal Policy

Two categories of Fiscal Policy

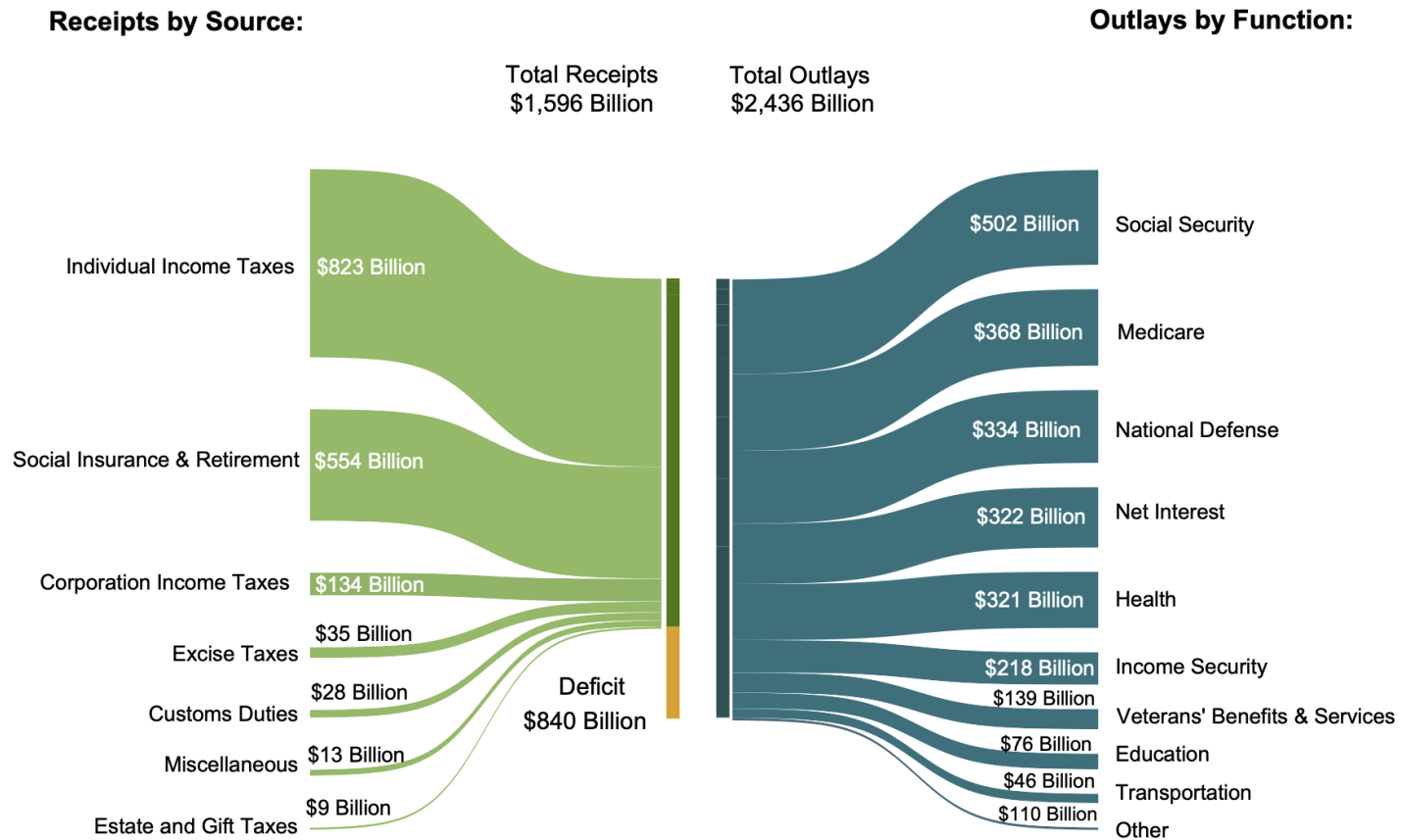
1. Discretionary Fiscal Policy

2. Automatic Stabilizers

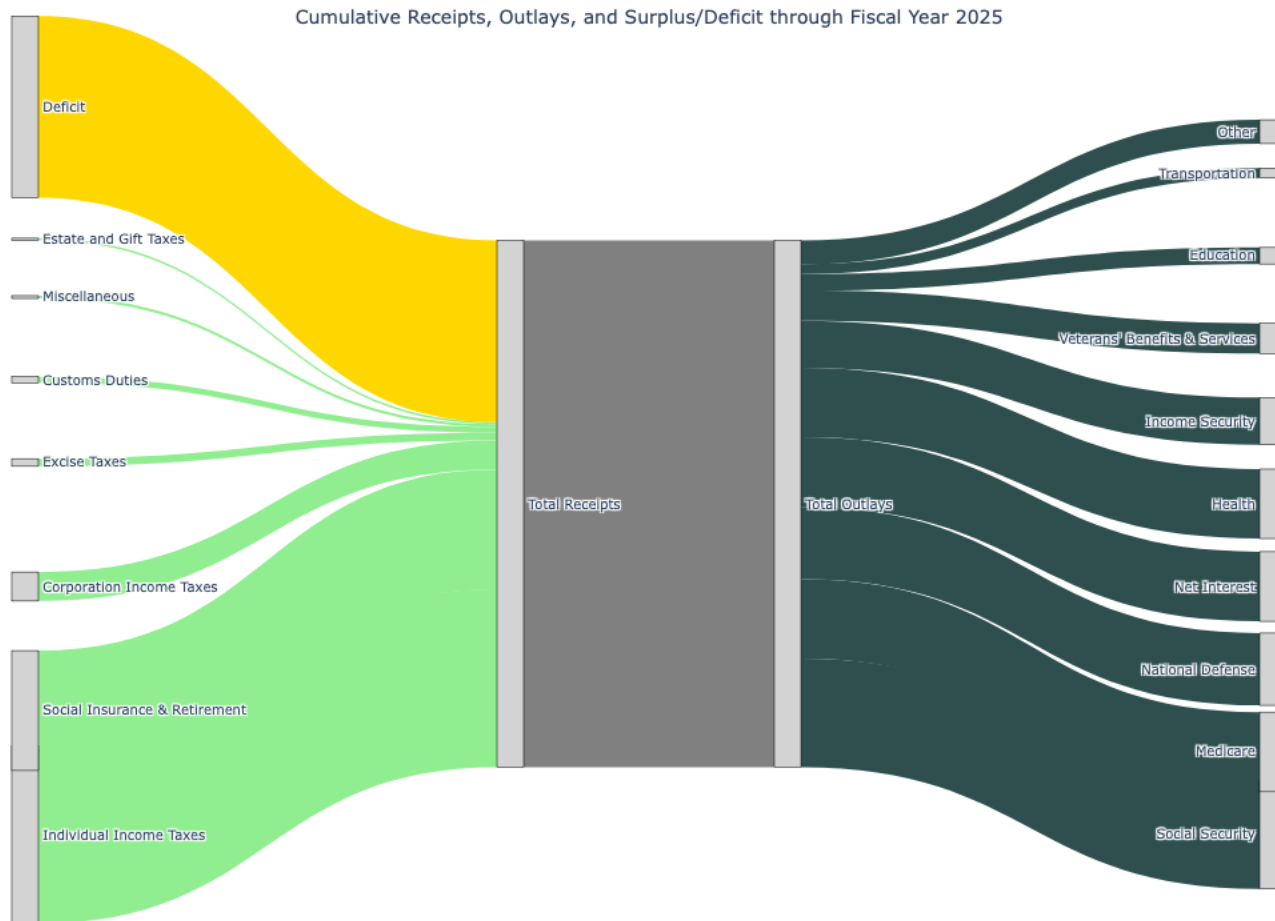


Federal Government Expenditures (since 10/2024)

Figure 2. Cumulative Receipts, Outlays, and Surplus/Deficit through Fiscal Year 2025

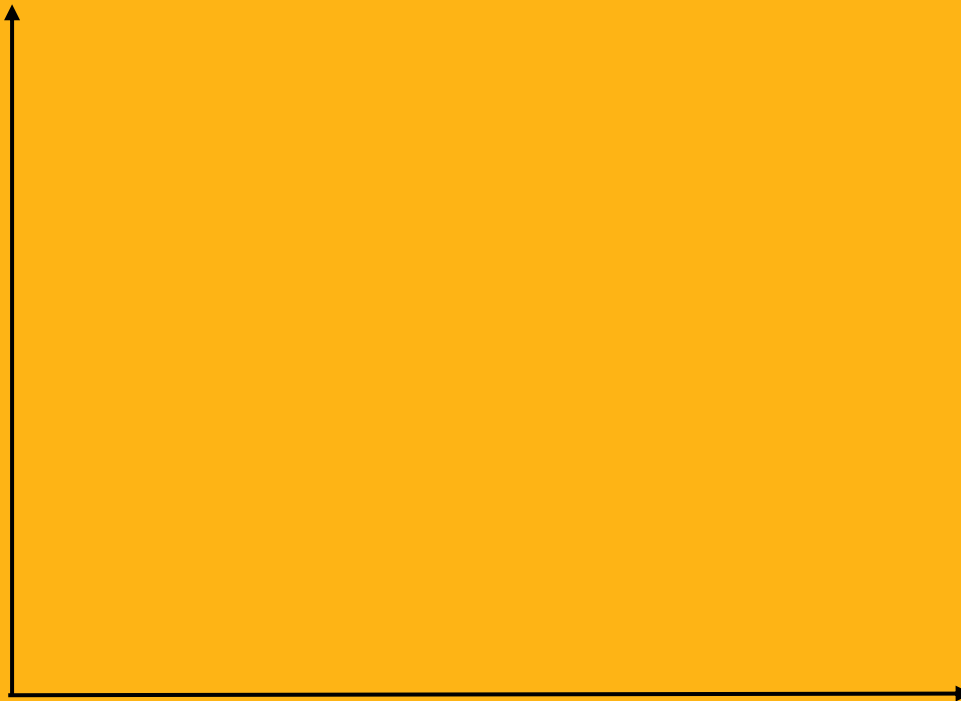


Federal Government Revenue



How can the Government use Fiscal Policy?

- Y^* decreases to \$7,500
- What should the government do to bring output up to \$10,000 but **mpc=0.90**?
- Option 1: Increase Government Purchases (same as before)



How can the Government use Fiscal Policy?

- Option 1: Increase G by \$250, $mpc=0.9$

Round 1:
Y up \$250

\$250 in new bridge = increase G by \$250
Incomes increase by \$250

Round 2:
Y up \$225 more,
\$475 total

Spend $mpc \times 250 =$
 $0.90 \times 250 = \$225$
Consumption increase

Save $mps \times 250 =$
 $0.10 \times 250 = \$25$

Incomes rise by \$225

Round 3:
Y up \$202.50,
\$677.5 total

Spend $mpc \times 225 =$
 $0.90 \times 225 = \$202.5$
Consumption increase

Save $mps \times 225 =$
 $0.10 \times 225 = \$22.50$



Changing Government Purchases or Taxes?



- How does the government increase G ?
- Congress **AGREES** on a change in G ! President does not veto a change in G !
- Example: A new bridge is built over the Ohio River in Cincinnati on I-71/75, $\Delta G = 250$



Problems with Changing G

1. Time Lag: Can Congress respond quickly?

2. Implementing Spending: Who builds the bridge? When?

3. Irreversibility: What if economy recovers before completion of the project?



How can the Government use Fiscal Policy?

Option 2: Decrease Taxes by \$250, $mpc = 0.9$

Decreasing T misses round 1 increase in Y , but all other rounds are the same.

Increasing G by \$250 increased Y by \$2,500

Decreasing T by \$250 will increase Y by less than \$2,500. How much less?

Tax Multiplier:



How can the Government use Fiscal Policy?

- Option 2: Decrease Taxes by \$250, $mpc = 0.9$

Round 1:
No change in Y
Disposable Y up
\$250

\$250 in tax breaks to workers = decrease T by \$250
Change in Y? None

Round 2:
Y up \$225 more,
\$225 total

Spend $mpc \times 250 =$
 $0.90 \times 250 = \$225$
Consumption increase

Save $mps \times 250 =$
 $0.10 \times 250 = \$25$

Incomes rise by \$225

Round 3:
Y up \$202.50,
\$427.5 total

Spend $mpc \times 225 =$
 $0.90 \times 225 = \$202.5$
Consumption increase

Save $mps \times 225 =$
 $0.10 \times 225 = \$22.50$

Tax Multiplier

What is the value of the tax multiplier?

For every \$1 increase in taxes, Y will change by...

For every \$1 decrease in taxes, Y will change by...



Problems with Tax Cuts

Simple Option 1: Cut taxes for low income households

Benefits:

Costs:



Problems with Tax Cuts



Simple Option 2: Cut taxes for rich

Benefits: \$1 million dollar salary = \$353,168.19
in Federal taxes

Costs:

Forward-Looking Behavior: Tax cuts today
mean tax increases in the future

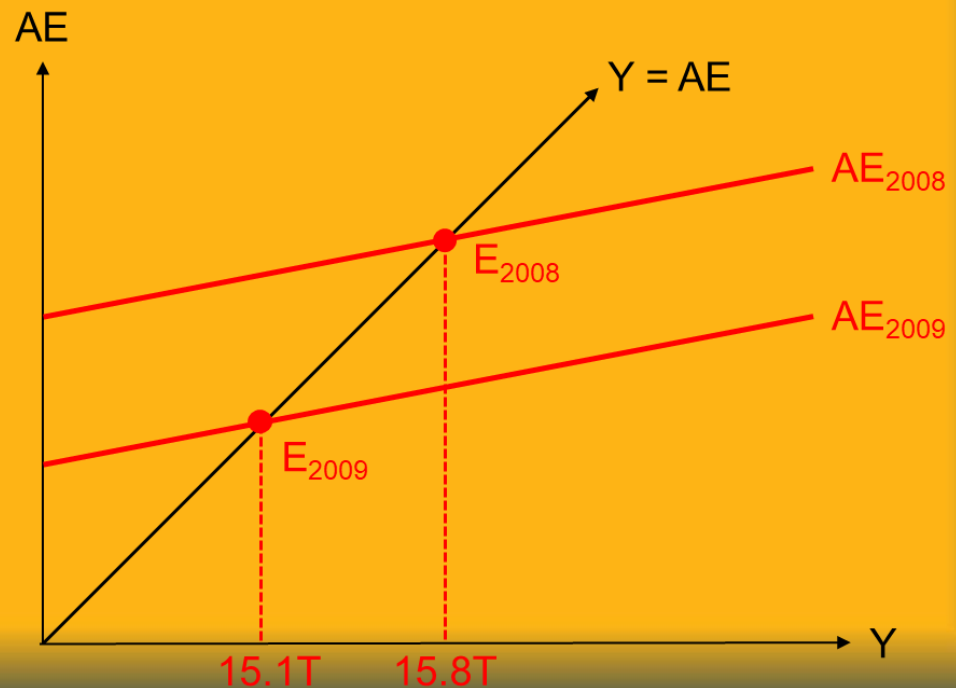
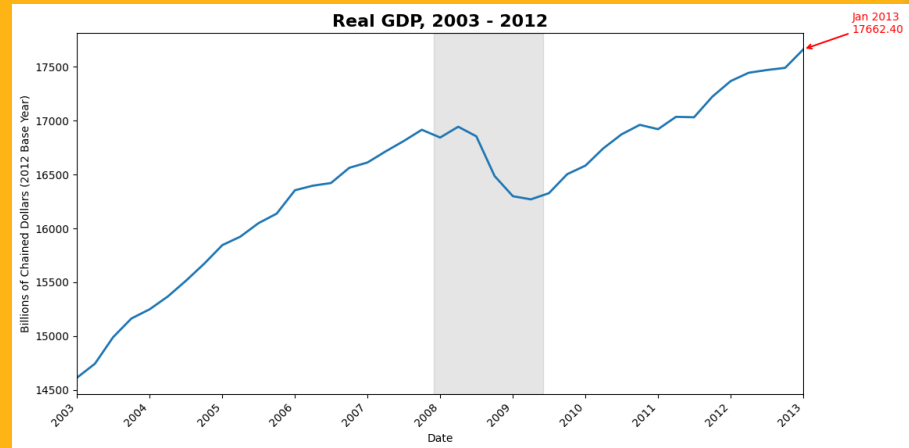
Federal Reserve Response?

Biggest Concern with Fiscal Policy?

Government Response to the Great Recession

Economic crises = fiscal policies

Great Financial Crisis
Covid-19 pandemic



Government Response to the Great Recession

2008: Troubled Asset Relief Program (TARP) = Bank Bailout Bill

\$700 billion in “troubled assets” authorized to be bought by the US Treasury

Government bought stocks in banks: Goldman Sachs, JP Morgan, Wells Fargo, etc.

Banks required to pay back loans with a 5% dividend (up to 9% in 2013)

Loans to:

American Insurance Group (\$40 billion)

Community Banks (\$92 billion)

Big 3 Auto Companies (\$25 billion)

Citigroup and Bank of American (\$45 billion)

Only \$350 billion of \$700 billion was used to stabilize the financial markets

Concerns about why certain banks were “bailed out” (discrimination)

US Treasury actually made a \$15 billion profit from TARP

Government Response to the Great Recession

2009: American Recovery and Reinvestment Act

\$800+ billion package of tax cuts and government spending increases

Tax Cuts = \$300 billion, $\Delta T = -\$300$ billion

State and Local Government Spending = \$150 billion, $\Delta G_1 = \$150$ billion

Federal Government Spending = \$350 billion, $\Delta G_1 = \$350$ billion

How much will Y change? Depends on mpc!

Assume the mpc = 0.5

Tax Multiplier =

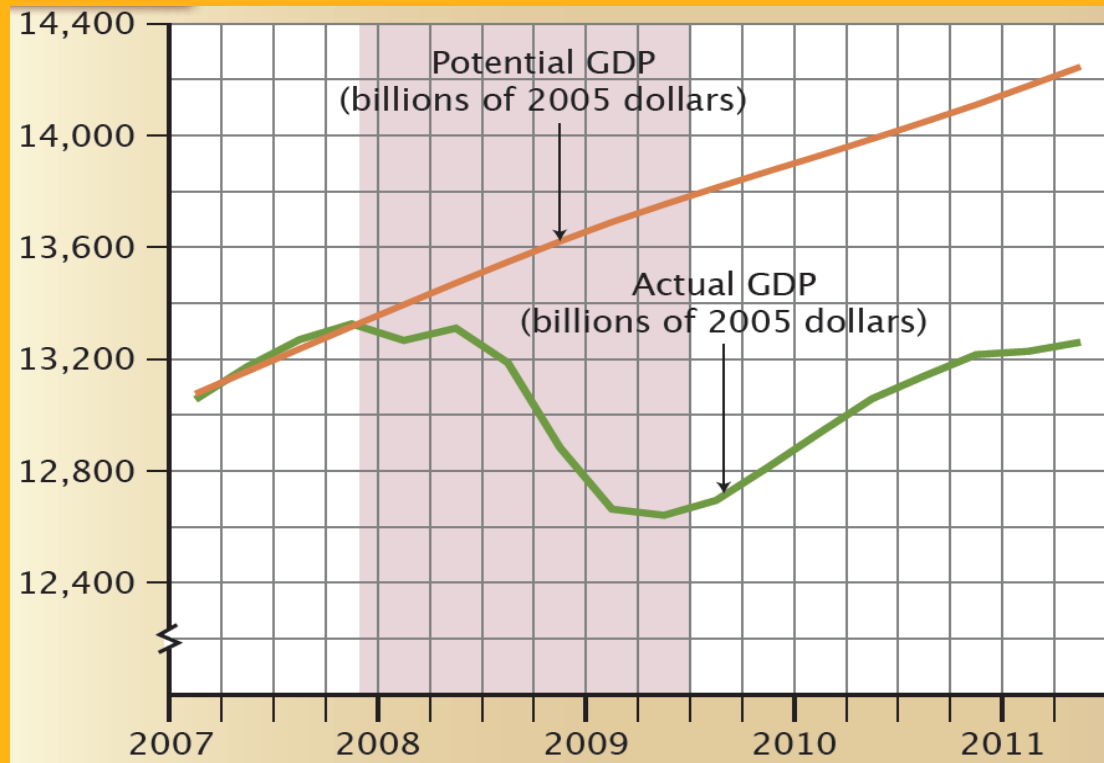
Expenditure Multiplier =

Total Increase in Y after 2009?



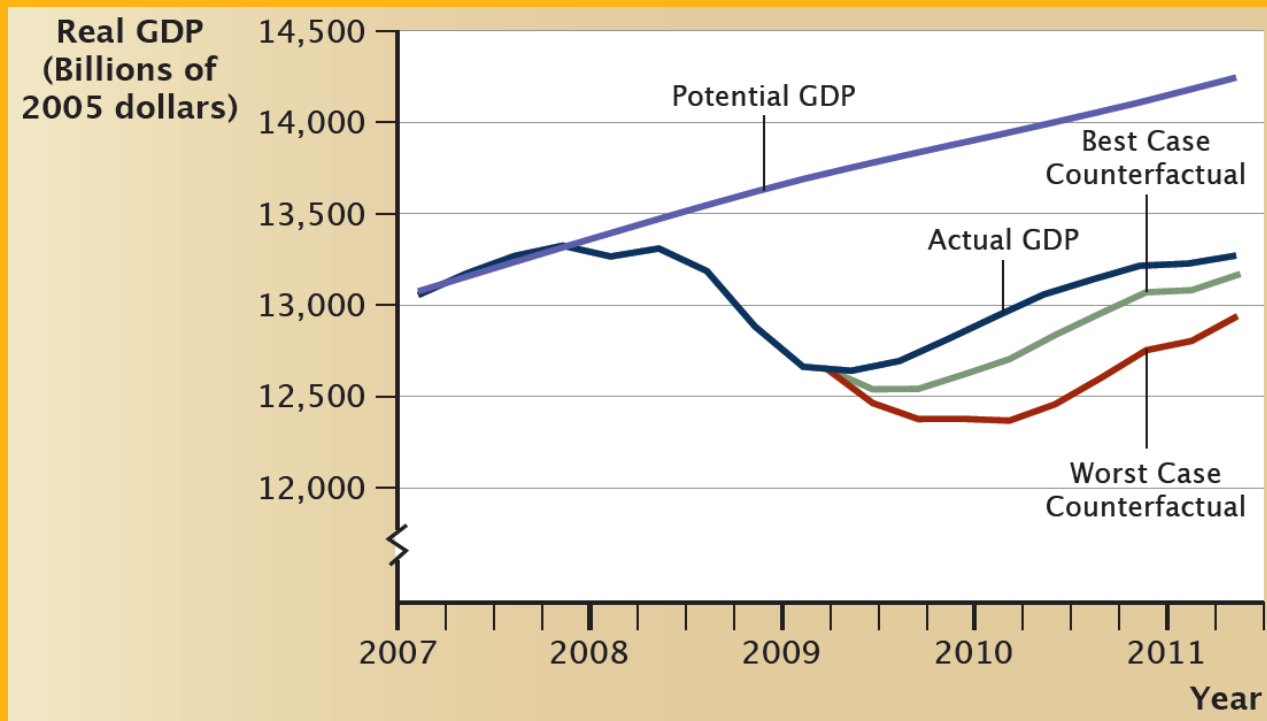
Government Response to the Great Recession

- Was stimulus effective?



Government Response to the Great Recession

- Was stimulus useless?



Government Response to the Great Recession

Purchases of goods and services by the federal government	1.0 to 2.5
Transfers to state and local governments for infrastructure	0.5 to 2.5
Transfers to state and local governments for other purposes	0.7 to 1.8
Transfer payments to individuals (such as unemployment benefits and food stamps)	0.8 to 2.1
One-time payments to retirees and disabled individuals	0.3 to 1.0
One-time payments to existing recipients of Social Security	0.3 to 1.0
Making Work Pay tax credit	0.5 to 1.7
AMT patch	0.2 to 0.6
Other individual tax provisions	0.3 to 1.5

Government Response to the Great Recession

The size of the Great Recession stimulus package is still debated

Some say it was too small to stimulate the economy

Others claim it was too large and led to inflation and debt

Problems with the package's design besides its size

